

BUSINESS CASE

Sinopec Supply-Chain Optimization Investment Project

Purpose: Executive investment decision

Customer: Sinopec

Initiative: Crude-oil supply-chain planning and optimization

Content:

- Executive Summary
- Business Need / Current State
- Case for Change
- Business Objectives
- Value Driver Tree
- Benefits Model
- Investment / TCO Model
- Five-Year Benefit Ramp
- Cash-Flow Model
- Financial Evaluation
- Sensitivity Analysis
- Risks Analysis
- Implementation / Value Realization Plan
- Executive Decision Page

1. Executive Summary

1.1 Business challenge

Sinopec's crude-oil planning process was highly complex:

- More than **200 variables**
- More than **200 people involved**
- Approximately **3 weeks** for the monthly planning cycle

This created a significant opportunity to improve the speed and economics of procurement planning.

1.2 Proposed solution

Deploy a unified supply-chain optimization platform capable of integrating the relevant planning variables, fragmented planning activities crossing the entire organization, and supporting more economically optimized crude-oil purchasing decisions.

1.3 Expected business value

The business case focuses on three potential value drivers:

Value driver	Annual Value Generation
Crude-oil procurement basket optimization	\$15.5M
Planning productivity	\$7.0M
Downstream economic optimization	\$7.7M
Total annual benefit	\$30M

1.4 Investment

Investment component	Amount
Implementation and modeling	\$700k
Software license	\$17.5M
Model maintenance	\$6.4M

Investment component	Amount
Server, network and other hardware	\$110k
5-year costs	\$24M
5-year TCO	\$24.7M

1.5 Financial outcome

Using a 9.0% discount rate (given by customer) and the modeled benefit ramp:

- **5-year gross benefits:** \$143M
- **5-year net benefits:** \$119M
- **ROI:** 500%
- **NPV:** \$90M
- **IRR:** 1483%
- **Payback:** approximately within the first year (12 months)

1.6 Strategic conclusion

This business case demonstrates how a relatively complex operational problem can potentially generate substantial economic value when translated into a quantified investment case.

2. Business Need / Current State

2.1 Current State:

200+ variables → 200+ people → 3-week planning cycle for monthly planning → slower decision-making & high oil purchasing expenditure → economic optimization opportunity

2.2 Business Need:

Sinopec needs to make complex crude-oil procurement decisions with most optimal purchasing plan quickly enough and accurately enough to optimize economic outcomes.

3. Case for Change

3.1 Current state

Dimension	Current state
Planning variables	200+
People involved	200+
Planning cycle	3 weeks
Planning environment	Highly complex
Economic decision	Crude procurement

3.2 Desired state

Dimension	Target
Planning cycle	<1 week
Decision-making	Faster and more economically optimized
Organizational adoption	Enterprise-scale
Business outcome	Procurement cost optimization

3.3 Change for Business Impact:

Operational problem → Economic consequence → Investment opportunity

4. Business Objectives

4.1 Primary objectives

- a) Reduce crude-oil planning cycle time.
- b) Improve the economics of crude procurement decisions.
- c) Find most optimal purchasing planning crossing the 200+ variables.
- d) Enable broader organizational adoption of optimized planning.
- e) Establish measurable and sustainable value realization.

4.2 Success measures

KPIs include:

- Planning-cycle duration
- Procurement cost
- Maximized gain from selling high-value products in downstream market
- Optimization opportunity captured
- Number of planning scenarios evaluated
- User adoption
- Number of participating business units
- Realized savings versus business-case baseline

5. Value Driver Tree

5.1 Business problem

Complex and slow crude-oil planning

↓

5.2 Value drivers

1) Procurement optimization

Better crude specification selection and purchasing decisions.

↓

2) Planning productivity

Reduce time and effort required to evaluate scenarios and develop plans.

↓

3) Downstream economic optimization

Potentially improve the economic consequences of crude selection on downstream operations and high-value product economics.

↓

5.3 Economic impact

Lower procurement cost + productivity benefit + downstream economic benefit

↓

5.4 Customer value

~\$30M annual reported procurement savings

6. Benefits Model

Benefit	Calculation logic	5-year value
Crude procurement optimization	Economic improvement from optimized crude purchasing	\$15.5M
Planning productivity	Reduced planning effort/time	\$7.0M
Downstream economics	Potential incremental economic contribution from selling more high-value products	\$7.7M

Benefit	Calculation logic	5-year value
Total		\$30M

7. Investment / TCO Model

7.1 Initial investment

Cost	Year 0
Implementation/deployment	\$700k

7.2 5-year costs

Cost	5 Years
Software license	\$17.5M
Model maintenance	\$6.4M
Server, network and other hardware	\$110k
5-Year Cost	\$24M

7.3 TCO (Over five years):

\$700kM initial + \$24M = \$24.7M TCO

8. Five-Year Benefit Ramp

8.1 Benefit realization:

Year	Benefit realization	Gross benefit
Year 1	46%	\$13.7M
Year 2	100%	\$31M
Year 3	100%	\$32M
Year 4	100%	\$33M
Year 5	100%	\$34M
Total		\$143.7M

9. Cash-Flow Model

Year	Gross benefit	Cost	Net cash flow
Year 0	\$0M	\$700k	-\$700k
Year 1	\$13.7M	\$4.5M	\$9.3M
Year 2	\$31M	\$4.6M	\$26.4M
Year 3	\$32M	\$4.8M	\$27.1M
Year 4	\$33M	\$4.9M	\$28.0M
Year 5	\$34M	\$5.0M	\$28.9M

10. Financial Evaluation

Using 9.0% discount rate given by customer (WACC):

10.1 ROI

- 1) Five-year gross benefit:

\$143M

- 2) Five-year TCO:

\$24M

- 3) Net benefit:

\$119M

- 4) ROI:

$\$123\text{M} \div \$15\text{M} \approx 500\%$

10.2 NPV

Discounting the future cash flows at 9.0%:

\$90M

10.3 IRR

1483%

10.4 Payback

12 Months.

11. Sensitivity Analysis

Scenario	Annual benefit	Remarks
Conservative	\$20M	Brent oil price as low as \$30/barrels
Base	\$29M	Strong modeled economic case, 95% possibility
Upside	\$36M	Brent oil price higher than \$160/barrels, advanced modeling

12. Risk Analysis

Risk	Potential impact	Mitigation
Adoption slower than expected	Benefit realization delayed	Phased rollout and adoption KPIs
Data quality	Optimization accuracy affected	Data-quality assessment
Model input change	Economic outcome changes	Scenario/sensitivity analysis
Organizational resistance	Adoption constrained	Executive sponsorship and change management
Benefit attribution	Savings difficult to verify	Establish baseline and benefit-tracking methodology

13. Implementation / Value Realization Plan

Phase 1 — Baseline

Establish:

- Current planning cycle
- Current procurement economics
- Current resource requirements
- Current KPIs

Phase 2 — Deploy

Implement the optimization capability and establish the operating model.

Phase 3 — Adopt

Expand usage across business units.

52 teams → 150+ active users

Phase 4 — Realize

Track:

- Planning-cycle reduction
- Procurement economics
- Adoption
- Actual savings
- Variance against business-case assumptions

Phase 5 — Optimize

Compare:

Business-case expectation vs. actual realized value

14. Executive Decision Page

14.1 The opportunity

Sinopec's crude-oil planning process involved **200+ variables, 200+ people and approximately three weeks per planning cycle**, creating a significant opportunity to improve procurement economics.

14.2 The value

~\$28M annual procurement savings with 95% possibility with:

- weeks → <1 week planning cycle
- and enterprise adoption across:
- 52 business units / 150+ active users

14.3 The investment

5-year TCO: \$24M

14.4 Modeled economics

- \$143M gross benefits
- \$119M net benefits
- 500% ROI
- \$90M NPV
- 12-month payback

14.5 Key decision considerations

- 1) Validated benefit baseline.
- 2) Validated investment assumptions.
- 3) Validated data inputs
- 4) Confirm implementation feasibility.
- 5) Established executive ownership of value realization.
- 6) Agreed on KPIs and benefits-tracking methodology.